EXPERIMENT-1

AIM: - Study of various Network commands used in Linux and Windows:

Basic Networking Commands

Command	Description
`arp -a`	ARP is short form of address resolution protocol. It will show the IP address of your computer along with the IP address and MAC address of your router.
`hostname`	This is the simplest of all TCP/IP commands. It simply displays the name of your computer.
`ipconfig /all`	This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.
`nbtstat -a`	This command helps solve problems with NetBIOS name resolution. (Nbt stands for NetBIOS over TCP/IP)
`netstat`	(network statistics) netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc. e.g.- `netstat -r`
`nslookup`	(name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in two modes: interactive and non-interactive. e.g.- `nslookup www.google.com`
`pathping`	Pathping is unique to Window's, and is basically a combination of the Ping and Tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.
`ping`	(Packet INternet Groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to other devices. 1. `#ping hostname( ping localhost)` 2. `#ping ip address (ping 4.2.2.2)` 3. `#ping fully qualified domain name(ping www.facebook.com)`
`route`	route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Some Important Linux Networking Commands

1. ip

The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems. The ip command can show address information, manipulate routing, plus display network various devices, interfaces, and tunnels.

Syntax:

```
ip <OPTIONS> <OBJECT> <COMMAND>
```

a. To show the IP addresses assigned to an interface on your server:

```
[root@server ~]# ip address show
```

b. To assign an IP to an interface, for example, enps03:

```
[root@server ~]# ip address add 192.168.1.254/24 dev enps03
```

c. To delete an IP on an interface:

```
[root@server ~]# ip address del 192.168.1.254/24 dev enps03
```

d. Alter the status of the interface by bringing the interface **eth0** online:

```
[root@server ~]# ip link set eth0 up
```

e. Alter the status of the interface by bringing the interface **eth0** offline:

```
[root@server ~]# ip link set eth0 down
```

f. Alter the status of the interface by enabling promiscuous mode for eth0:

```
[root@server ~]# ip link set eth0 promisc on
```

g. Add a default route (for all addresses) via the local gateway 192.168.1.254 that can be reached ondevice **eth0**:

```
[root@server ~]# ip route add default via 192.168.1.254 dev eth0
```

h. Add a route to 192.168.1.0/24 via the gateway at 192.168.1.254:

```
[root@server ~]# ip route add 192.168.1.0/24 via 192.168.1.254
```

i. Add a route to 192.168.1.0/24 that can be reached on device **eth0**:

```
[root@server ~]# ip route add 192.168.1.0/24 dev eth0
```

j. Delete the route for 192.168.1.0/24 via the gateway at 192.168.1.254:

```
[root@server ~]# ip route delete 192.168.1.0/24 via 192.168.1.254
```

2. ifconfig

The ifconfig command was a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed above.

3. mtr

MTR (Matt's traceroute) is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute commands. Just like a traceroute, the mtr command will show the route from a computer to a specified host. mtr provides a lot of statistics about each hop, such as response time and percentage. With the mtr command, you will get more information about the route and be able to see problematic devices along the way. If you see a sudden increase in response time or packet loss, then obviously, there is a bad link somewhere. The syntax of the command is as follows:

```
mtr <options> hostname/IP
```

Let's look at some common use cases.

a. The basic mtr command shows you the statistics, including each hop (hostnames) with time andloss%:

```
[root@server ~]# mtr google.com
```

b. Show numeric IP addresses (if you use -g, you will get IP addresses (numbers) instead ofhostnames):

```
[root@server ~]# mtr -g google.com
```

c. Show the numeric IP addresses and hostnames, too:

```
[root@server ~]# mtr -b google.com
```

d. Set the number of pings that you want to send:

```
[root@server ~]# mtr -c 10 google.com
```

4. tcpdump

The tcpdump command is designed for capturing and displaying packets.

You can install tcpdump with the command below:

```
[root@server ~]# dnf install -y tcpdump
```

Before starting any capture, you need to know which interfaces tcpdump can use. You will need to use sudo or have root access in this case.

```
[root@server ~]# tcpdump -D
```

If you want to capture traffic on eth0, you can initiate that with `tcpdump -i eth0`

```
[root@server ~]# tcpdump -i eth0
[root@server ~]# tcpdump -i eth0 -c 10
```

Capture traffic to and from one host

You can filter out traffic coming from a specific host. For example, to find traffic coming from and going to 8.8.8.8, use the command:

```
[root@server ~]# tcpdump -i eth0 -c 10 host 8.8.8.8
```

For traffic coming from 8.8.8.8, use:

```
[root@server ~]# tcpdump -i eth0 src host 8.8.8.8
```

For outbound traffic going to 8.8.8.8, use:

```
[root@server ~]# tcpdump -i eth0 dst host 8.8.8.8
```

Capture traffic to and from a network

You can also capture traffic to and from a specific network using the command below:

```
[root@server ~]# tcpdump -i eth0 net 10.1.0.0 mask 255.255.255.0
```

or:

```
[root@server ~]# tcpdump -i eth0 net 10.1.0.0/24
```

Capture traffic to and from port numbers

Capture only DNS port 53 traffic:

```
[root@server ~]# tcpdump -i eth0 port 53
```

For a specific host,

```
[root@server ~]# tcpdump -i eth0 host 8.8.8.8 and port 53
```

To capture only HTTPS traffic,

```
[root@server ~]# tcpdump -i eth0 -c 10 host www.google.com and port 443
```

To capture all port except port 80 and 25,

```
[root@server ~]# tcpdump -i eth0 port not 53 and not 25
```

5. ping

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages. The receipt of corresponding Echo Reply messages is displayed, along with round-trip times. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability, and name resolution.